

In the following exercises, all coordinates and components are given with respect to a right-oriented orthonormal frame $\mathcal{K}$.

8.1. Using the Euler-Rodrigues formula, deduce the known matrix forms of rotations around the coordinate axes.

8.2. Consider the affine transformation

$$\phi(\mathbf{x}) = \frac{1}{3} \begin{bmatrix} -1 & 2 & -2 \\ -2 & -2 & -1 \\ -2 & 1 & 2 \end{bmatrix} \mathbf{x} + \begin{bmatrix} \lambda \\ 0 \\ 1 \end{bmatrix}.$$

Explain why it is an isometry. Discuss the type of the isometry in terms of the parameter λ .

8.3. Show that a matrix $A \in O(3)$ of determinant -1 admits -1 as an eigenvalue.

8.4. For a matrix A and a vector $\mathbf{v}$ such that $A\mathbf{v} = -\mathbf{v}$, show that the plane

$$\pi : \langle \mathbf{v}, \mathbf{x} \rangle = \frac{1}{2} \langle \mathbf{v}, \mathbf{b} \rangle$$

is invariant under the isometry $\phi(\mathbf{x}) = A\mathbf{x} + \mathbf{b}$.

8.5. Using rotations around the coordinate axes, give a parametrization of a cone containing the line $\ell = \{(0, t, t) : t \in \mathbb{R}\}$ and with axis the z -axis.

8.6. Using the Euler-Rodrigues formula, write down the matrix form of a rotation around the axis $\mathbb{R}\mathbf{v}$ where $\mathbf{v} = (1, 1, 0)$. Use this matrix form to give a parametrization of a cylinder with axis $\mathbb{R}\mathbf{v}$ and diameter $\sqrt{2}$.

8.7 (Paraboloid of revolution). Give an equation for the locus of points $\mathcal{P}$ which are equidistant from the xy -plane and the point $(0, 0, 1)$. Give a parametrization of the surface $\mathcal{P}'$ obtained by rotating the parabola $z = \frac{1}{2}x^2 + \frac{1}{2}$ from the xz -plane around the z -axis. Show that $\mathcal{P} = \mathcal{P}'$.

8.8. Discuss, in dimension 3, the possible isometries obtained by composing a reflection in a plane π_1 with a reflection in a plane π_2 in terms of the relative positions of the two planes.